

Biodiversity and its Conservation

Biodiversity refers to the variety and variability among all groups of living organisms and the ecosystem complexes in which they occur.

- **In the Convention of Biological diversity (1992) held in Rio de Janeiro biodiversity has been defined as the variability among living organisms from all sources including inter alia, terrestrial, marine and other aquatic ecosystems and the ecological complexes of which they are a part.**

SPECIES DIVERSITY

- This is the variability found within the population of a species or between different species of a community.
- It represents broadly the species richness and their abundance in a community. There are two popular indices of measuring species diversity known as **Shannon-Wiener index** and **Simpson index**

- The Shannon diversity index (a.k.a. the Shannon–Wiener diversity index) tells you how diverse the species in a given community are. It rises with the number of species and the evenness of their abundance – Randomness Index
- Tells us about the Richness and Evenness of the species, e.g., rich of nightingales
- Simpson's Diversity Index is a measure of diversity which takes into account the number of species present, as well as the relative abundance of each species - Dominance index. Crows dominate in this forest.

- **biological Diversity – Genetic, Ecosystem and more importantly Species Diversity.**
- **The Shannon-Wiener Species Diversity Index is calculated by taking the number of each species, the proportion each species is of the total number of individuals, and sums the proportion times the natural log of the proportion for each species.**
- **Simpson's Diversity Index is used to calculate a measure of diversity, taking into account the number of something as well as its abundance.**

Radial Analysis-Remains of Animal Bones- Food Access (Green cover) and Consumption

- **Motion-sensor cameras, Water proof cameras and Satellite imagery etc. are employed.**
- **The Shannon-Wiener Diversity Index, H , is calculated using the following equation: $H = -\sum P_i (\ln P_i)$ where P_i is the proportion of each species in the sample. Given a very large sample size, with more than 5 species, the S-W index values (H) can range of 0 to ~ 4.6 using the natural log ($\ln$).**
- **(π)**

Species Richness and Relative Abundance

- Example
- Problem Statement:
- The samples of 5 species are 60,10,25,1,4. Calculate the Shannon diversity index and Evenness for these sample values.
- Sample Values (S) = 60,10,25,1,4 number of species (N) = 5
- First, let us calculate the sum of the given values.
- Sum = $(60+10+25+1+4) = 100$

Example

- Species (i)

- | | No. in sample | p_i | $\ln(p_i)$ | $p_i \times \ln(p_i)$ |
|---------------|---------------|-------|------------|-----------------------|
| Big bluestem | 60 | 0.60 | -0.51 | -0.31 |
| Partridge pea | 10 | 0.10 | -2.30 | -0.23 |
| Sumac | 25 | 0.25 | -1.39 | -0.35 |
| Sedge | 1 | 0.01 | -4.61 | -0.05 |
| Lespedeza | 4 | 0.04 | -3.22 | -0.13 |
- $S = 5$, Sum = 100 Sum = -1.07
 - $H = 1.07$, $H_{\max} = \ln(S) = \ln(5) = 1.61$,
 - $E = H/H_{\max} = 1.07/1.61 = 0.66$ i.e., Evenness (E) = 0.66
 - $H = 1.07$ i.e., Shannon diversity index (H) = 1.07
 - Good Shannon-Wiener index value is between 1.5 and 3.5.

Calculation of Simpson's Index

- **Simpson's Diversity Index Calculation**

1. Add the individual species populations to get N .

2. Determine $N \times (N - 1)$.

3. Where n is the number of individuals in each species. Work out $n \times (n - 1)$ for each species,

4. Sum all the values in step 3.

5. Divide the sum obtained in step 4 by the value obtained in step 2.

$$D = 1 - \left[\frac{n(n-1)}{N(N-1)} \right]$$

Simpson's Diversity Index = $1 - D$.

Simpson's diversity index- How many different types.

- **Sum all the values in step 3.**
- **Divide the sum obtained in step 4 by the value obtained in step 2. As a result, you will get Simpson's index D .**
- **Evaluate Simpson's diversity index as $1 - D$.**

Assignment 1

- Rare earth metals-Importance, uses, occurrence, production and current demand for permanent magnets.
- (Last date for submission : September 12, 2026) .
- Page limit is 5 - 6 pages , excluding figures and tables if any.
- Double spaced Font size 14.
- They may include some references which may not be counted within 5-6 pages. You may refer to the Internet, Books, Encyclopedia, Wikipedia etc. for the information. Write in your own words.
- Use of turnitin for checking plagiarism and deduction of marks for late submission and plagiarism. Institute rules may be followed.

ECOSYSTEM DIVERSITY

- **An ecosystem is a community of living things—like plants, animals, and germs—that interact with each other and their non-living surroundings, such as air, water, and soil, working together as a single, connected unit.**
- **This is the diversity of ecological complexity showing variations in ecological niches, trophic structure, food-webs, nutrient cycling etc.**
- **The ecosystems also show variations with respect to physical parameters like moisture, temperature, altitude, precipitation etc.**

Saving the Humanity from Emerging Disasters and Dangers

- The diversity has developed over millions of years of evolution. **If we destroy this diversity, it would disrupt the ecological balance and ultimately natural balance and of course our Environment where we live and we do not wish to experience or invite any problems such as,, dieases or other natural disasters and may finally result in the elimination of Human Life on this Terrestrial Ball like dinosaurs.**
- These are mainly due to variations in the physical factors such as moisture, temperature, altitude, precipitation, environmental pollution of air, water, soil, food etc. etc.

CAUSES OF BIODIVERSITY LOSS

- Climate change- Temperature, Storms, Earth Quakes etc.
- Pollution.
- Destruction of habitats.
- Invasive alien species.
- Overexploitation of the natural environment.
- Hunting
- Habitat Fragmentation
- Collection for Zoo and Research
- Control of Pests and Predators
- Genetic Mutations
- Co-extinction – When related or dependent species vanishes.

VALUE OF BIODIVERSITY

- The value of biodiversity in terms of its commercial utility, ecological services, social and aesthetic value is enormous.

1. Consumptive use value:

- These are direct use values where the biodiversity product can be harvested and consumed directly e.g. fuel, food, drugs, fibre etc.

Medicines derived from the natural products

- **Drugs and medicines:** About 75% of the world's population depends upon plants or plant extracts for medicines.
- The wonder drug Penicillin used as an antibiotics is derived from fungus called **Penicillium**. Likewise, we get Tetracyclin from a bacterium. **Quinine**, the cure for malaria is obtained from the bark of Cinchona tree. Artemisinin, taxol (Anticancer), Atropine etc.

Morphine and Codeine: Natural Products

- Digitalin is obtained from foxglove (*Digitalis*) plant which is an effective cure for **heart ailments**. Recently vinblastin and vincristine, two **anti-cancer drugs**, have been obtained from Periwinkle (*Catharanthus*) plant, which possesses anti-cancer alkaloids. A large number of marine animals/algae etc. are supposed to possess **anti-cancer properties** which are yet to be explored systematically.

Fuel:

Our forests have been used since ages for fuel wood. The fossil fuels coal, petroleum and natural gas are also products of fossilized biodiversity.

Firewood collected by individuals are not normally marketed, but are directly consumed by tribals and local villagers, hence falls under consumptive value.

2. Productive use values:

- These are the commercially usable values where the product is marketed and sold. It may include **lumber or wild gene resources** that can be traded for use by **scientists** for introducing desirable **traits in the crops and domesticated animals**.

- These may include the animal products like tusks of elephants, musk from musk deer, silk from silk-worm, wool from sheep, fur of many animals, lac from lac insects etc, all of which are traded in the market.
- Many industries are dependent upon the productive use values of biodiversity e.g.- the Paper and pulp industry, Plywood industry etc.
- Pulp – Delignification of wood.

- Railway sleeper industry, Silk industry, textile industry, ivory-works, leather industry, pearl Industry etc.
- Despite international ban on trade in products from endangered species, smuggling of fur, hide, horns, tusks, live specimen etc. worth millions of dollars are being sold every year.
- Developing countries in Asia, Africa and Latin America are the richest biodiversity centers and

Wild life products are smuggled and marketed in large quantities to some rich western countries and also to China and Hong Kong where export of cat skins and snake skins fetches a booming business.

Donkeys from Pakistan to China

3. Social Value: These are the values associated with the social life, customs, religion and psycho-spiritual aspects of the people. Many of the plants are considered holy and sacred in our country like Tulsi (holy basil), Peepal (Ficus) , Mango, Lotus, Bael Patthar (Stone apple) etc. The leaves, fruits or flowers of these plants are used in worship or the plant itself is worshipped. Refer to Ayurveda and Pharmacological properties.

- Their social life, songs, dances and customs are closely woven around the wildlife. **Many animals like Cow, Snake, Bull, Peacock, Owl etc. also have significant place in our psycho-spiritual arena and thus hold special social importance.** Thus biodiversity has distinct social value, attached with different societies.
- Snakes in Samastipur

Video on Nagpanchami festival in Bihar

- Snakes are very dangerous and we should be careful of them. And we should not touch them.
- <https://www.indiatoday.in/india/video/video-devotees-carry-snakes-in-bihars-samastipur-during-nag-panchami-fair-2757323-2025-07-17>

**Snakes are dangerous - we should be
very careful.**

Shetpal village in Maharashtra



इस गाँव में हर घर और गली में नाग (साँप) खुलेआम घूमते हैं। इस गाँव का नाम शेतफल है जो महाराष्ट्र के सोलापुर ज़िले में स्थित है। गाँववाले साँपों को परिवार का हिस्सा मानते हैं और उनसे डरते नहीं हैं। गाँव में श्री शेषनाग का एक प्राचीन मंदिर है और माना जाता है कि इस गाँव को नागदेवता (शेषनाग) का आशीर्वाद प्राप्त है।

रोचक जानकारी

4. Ethical value: It is also sometimes known as existence value. It involves ethical issues like “ all life must be preserved . It is based on the concept of

Live and Let Live

If we want our human race to survive, then we must protect all biodiversity, because biodiversity is valuable. – Interdependence.

Role of each species on this earth to keep it clean and in working condition by maintaining natural balance that has evolved over the millions to billions of years.

Ants are vital to the earth because they act as natural ecosystem engineers. They aerate soil through extensive tunneling, disperse seeds to help plants grow, control pest populations, and recycle organic waste by decomposing dead matter

- The ethical value means that we may or may not use a species, but knowing the very fact that this species exists in nature gives us pleasure. **We all feel sorry when we learn that passenger pigeon” or dodo is no more on this earth.**
- You may see the video of birds at Kruger National Park, South Africa.
- <https://www.youtube.com/watch?v=UQxZmCnl1Oc> , accessed on August 23, 2023.

Songs of Birds

- We are not deriving anything direct from Kangaroo, Zebra or Giraffe, but we all strongly feel that these species should exist in nature. This means, there is an **ethical value or existence value** attached to each species. People do not ride zebras because they are dangerously aggressive, unpredictable, and lack the social hierarchy or back strength needed for easy riding. Evolution shaped them to fight fierce predators like lions, meaning they bite hard and kick to kill, unlike horses which were successfully tamed and selectively bred over thousands of years. **Nightingale**, Cuckoo **and several colourful birds in the forests, zoos and gardens .**

People from far and wide spend a lot of time and money to visit wilderness areas where they can enjoy the aesthetic value of biodiversity and this type of tourism is now known as eco-tourism. The willingness to pay on such eco-tourism gives us even a monetary estimate for aesthetic value of biodiversity.

What are Safaris?

Ecotourism Market

- Ecotourism gives the aesthetic value of biodiversity.
- The global ecotourism market was valued at approximately USD 235 billion to USD 244 billion, while the India.

5. Option values: These values include the potentials of biodiversity that are presently unknown and need to be explored. There

is a possibility that we may have some potential cure for AIDS or cancer existing within the depths of a marine ecosystem, or a tropical rainforest - Natural Products, Bioinformatics, Computational Biology-CADD (Computer aided drug design) etc.

Medicines derived from snake venom treat conditions like high blood pressure and blood clots, while scorpion venom is being studied for pain relief and cancer treatments.

- **The biodiversity is like precious gifts of nature presented to us. We should not commit the folly of losing these gifts even before unwrapping them.**
- **The option value also includes the values, in terms of the option to visit areas where a variety of flora and fauna, or specifically some endemic, rare or endangered species exist.**
- **PROJECT WORK IS AN OPPORTUNITY.**

Ecosystem service value:

It refers to the services provided by ecosystems like prevention of soil erosion, prevention of floods, maintenance of soil fertility, cycling of nutrients, fixation of nitrogen, cycling of water, their role as carbon sinks, pollutant absorption and reduction of the threat of global warming **etc.**

- **Different categories of biodiversity value clearly indicate that ecosystem, species and genetic diversity all have enormous potential and a decline in biodiversity will lead to huge economic, ecological and socio-cultural losses.**
- **The decline of biodiversity poses severe dangers to ecosystem stability, human health, and global food security.**

GLOBAL BIODIVERSITY.

- **Following the 1992 The Earth Summit at Rio-de Janeiro, it became evident that there is a growing need to know and scientifically name, the huge number of species which are still unknown on this earth. Roughly 1.5 million species are known till date which is perhaps 1% or may be just 2% of the actual number.**

Tropical deforestation causes severe damage to global ecosystems, with scientists estimating that ongoing habitat destruction threatens a significant fraction of terrestrial species.

Tropical deforestation alone is reducing the biodiversity by half a percent every year.

Mapping biodiversity has become important.

Terrestrial Biodiversity.

This is described as Biomes.

Examples of Terrestrial Biodiversity are the tropical rainforests, tall grass prairies, savannas,/ grassland desert, tundra/treeless almost frozen etc.

Biome

This is known as a major life zone, is an area that includes communities of plants and animals that have a common adaptation to that particular environment.

This includes various communities which are named after the dominant features of that region – like grasslands, deserts or forests.

Tropical Rainforests

- The tropical rainforests are inhabited by teeming millions of species of plants, birds, amphibians (frogs, toads etc.), insects as well as mammals (having mammary glands). They are the Earth,s largest storehouses of biodiversity.
- About 50 to 80% of global biodiversity lies in these rainforests.

Floral Analysis – Seeds and nuts

- There is an estimated 1,25,000 flowering plant species in tropical forests. However, till now we know only 1-3% of these species.
- Several drugs are derived from plant species from here.

Temperate Forests

- Temperate forests have much less biodiversity, but there is much better documentation of the species.

Statistics May be deleted

Globally, we have roughly 1,70,000 flowering plants, 30,000 vertebrates and about 2,50,000 other groups of species that have been described.

- It is interesting to know that marine diversity is even much higher than terrestrial biodiversity and ironically, they are still less known and described.
- Out of the 35 existing phyla (categories) of multicellular animals, 34 are marine and 16 of these are exclusively marine.

BIOLOGICAL DIVERSITY AT NATIONAL LEVEL

(Indian Biodiversity)

India ranks 10th among the plant rich countries of the world, 11th in terms of number of endemic species of higher vertebrates and 6th among the centers of diversity and origin of agricultural crops.

The total number of living species identified in our country is 1,50,000.

REGIONAL OR LOCAL BIODIVERSITY

Biodiversity at regional level is better understood by categorizing species richness into three types, based upon their spatial distribution :

- (i) Point (α -) richness refers to the number of species that can be found at a single point in a given space.

Diversity on a local scale, such as a pond, a field, or a patch of forest.

(iii) Gamma (γ -) richness refers to the rate of change across large landscape gradients.

Gamma diversity is a measure of the overall diversity for the different ecosystems within a region e.g., species diversity along the entire range of the mountain slope.

(ii) Beta (β -) richness refers to the rate of change in species composition across different habitats.

It quantifies how species diversity changes from one habitat to another, providing insights into the spatial patterns of biodiversity i.e., between any 2 patches and their community.

INDIA AS A MEGA-DIVERSITY NATION

- Originally, beta diversity was defined as :
- β diversity = γ/α (sensu Whittaker, 1972).
- Additive formulation: β diversity = γ diversity - α diversity. (proposed by MacArthur et al. and Lande).

India is one of the 12 mega-diversity countries in the world.

- The Ministry of Environment and Forests, Climate Change, Govt. of India (2000) records 47,000 species of plants and 81,000 species of animals which is about 7% and 6.5% respectively of global flora and fauna.

Endemism:

- Species which are restricted only to a particular area are known as endemic. India shows a good number of endemic species.
- About 62% of amphibians (frogs, toads, newts, salamanders etc.) and 50% of lizards are endemic to India.
- Western ghats are the site of maximum endemism.

Center of origin - India

: A large number of species are known to have originated in India. Nearly 5000 species of flowering plants had their origin in India.

India has been the center of origin of 166 species of crop plants and 320 species of wild relatives of cultivated crops – lemon, onion, eggplant (brinjal), Indian gooseberry (called amla in Hindi), mango etc.

- **Marine diversity:** Along 7500 km long coastline of our country in the mangroves, estuaries, coral reefs, backwaters etc. there exists a rich biodiversity. More than 340 species of corals of the world are found here.
- The marine diversity is rich in mollusks, crustaceans (crabs etc.).
- Marine diversity spans from massive blue whales and social dolphins to vibrant coral reefs that support entire ocean.

HOT SPOTS OF BIODIVERSITY

- Areas which exhibit high species richness as well as high species endemism are termed as **hotspots of biodiversity**. The term was introduced by Myers (1988).
- There are 36 such hotspots of biodiversity on a global level out of which two are present in India, namely the **Eastern Himalayas and Western Ghats**.

- These hotspots covering less than 2% of the world's land area are found to have about 50% of the terrestrial biodiversity.
- According to Myers et al. (2000) an area is designated as a hotspot when it contains at least 0.5% of the plant species or 1500 vascular plants/ tube like tissues , as endemics.
- About 40% of terrestrial plants and 25% of vertebrate species (having spinal cord or backbone) are endemic and found in these hotspots.

- These hotspots lie in India extending into neighbouring countries namely, Indo-Burma region (covering Eastern Himalayas) and Western Ghats - Sri Lanka region.
- The Indian hotspots are not only rich in floral wealth and endemic species of plants but also reptiles, amphibians (that live in water as well as land) , swallow tailed butterflies and some mammals.

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- **Eastern Himalayas:** There are numerous deep and semi-isolated valleys in Sikkim which are extremely rich in endemic plant species. In an area of 7298 Km² of Sikkim about 4250 plant species are found of which 60% are endemic
- **Western Ghats:** It extends along a 17,000 Km² strip of forests in Maharashtra, Karnataka, Tamil Nadu and Kerala and has 40% of the total endemic plant species. 62% amphibians and 50% lizards are endemic to Western Ghats

THREATS TO BIODIVERSITY

- **Extinction or elimination of a species is a natural process of evolution.**
- **In the geologic period the earth has experienced mass extinctions.**
- **However, the rate of loss of species in geologic past has been a slow process, keeping in view the vast span of time going back to 444 million years.**
- **ALARMING SITUATION : The process of extinction has become particularly fast in the recent years of human civilization.**
- **Black-and-chestnut eagle, vulture, and Great Indian Bustard are facing severe threats pushing them toward extinction.**

Extinction

- One of the estimates by the noted ecologist, E.O. Wilson puts the figure of extinction at 10,000 species per year or 27 per day.
- If the present trend continues we would lose 1/3rd to 2/3rd of our current biodiversity by the middle of twenty first century.
- This would be very sad situation as this would disturb the balance of the nature and related natural phenomena which are keeping the mankind alive and ecosystems active on this earth.

SAVE PLANET – SAVE HUMANITY

- We may face extinction like dinosaurs and earlier human race on this very earth. We may become history.
- This is a Wake Up Call and that makes these studies at B.Tech level very urgent and important.

Video on Environmental Challenges on Earth

- https://www.google.com/search?q=environmental+issues+videos+for+students&rlz=1C1YTUH_enIN1045IN1045&sca_esv=f56ea74e923b65bc&biw=1507&bih=655&sxsrf=APpeQntLT1V8iNEtV03V1jcWRZuS94GUPA%3A1787391824130&ei=UG-JavLCB6CyhvcPmbyWsQQ&oq=&gs_lp=Egxnd3Mtd2l6LXNlcniAIACoCCAUYfxAAGIAEGIoFGJECGOcGGOOcGLQC2AEBMhcQABiABBiKBRiRAhjnBhjqaAhi0AtgBATIXEAAYgAQYigUYkQIY5wYY6gIYtALYAQEyFxAAGIAEGIoFGJECGOcGGOOcGLQC2AEBMhcQABiABBiKBRiRAhjnBhjqaAhi0AtgBATIXEAAYgAQYigUYkQIY5wYY6gIYtALYAQEyEBAAGAMYjwEY6gIYtALYAQEyEhAAGAMYjwEY6gIYtAIYCtgBATIQEAAYAXiPARjqAhi0AtgBATIQEAAYAXiPARjqAhi0AtgBAUiJaVAAWABwAXgBkAEAmAEAoAEAqgEAuAEByAEA-AEBmAIBoAIKqAIKmAAMK8QXBfCQUzuwibLoGBAgBGAEsBwExoAcAsgcAuAcAwgcDMY0xyAcIlgAgB&sclient=gws-wiz-serp#fpstate=ive&vld=cid:97e068ed,vid:QQYgCxu988s,st:0

LOSS OF HABITAT AND CAUSES

- Billions of hectares of forests and grasslands have been cleared over the past 10,000 years for conversion into agriculture lands, pastures, settlement areas or development projects.
- Currently solar panels and housing demands require land.
- The wetlands are destroyed due to draining, filling and pollution thereby causing huge biodiversity loss.

- *Sometimes the loss of habitat is in installments so that the habitat is divided into small and scattered patches, a phenomenon known as habitat fragmentation.*

- **POACHING**

Illegal trade of wildlife products by killing prohibited endangered animals i.e. poaching is another threat to wildlife.

Gila monster: Venom contains Exendin-4, a peptide that inspired modern GLP-1 receptor agonist medications for type 2 diabetes and weight management like Byetta and Ozempic. Cone snails: Venom produces Ziconotide, a potent non-opioid painkiller used for severe chronic pain management.

- Despite international ban on trade in products from endangered species, **smuggling of wildlife items like furs, hides, horns, tusks, live specimens and herbal products.** worth millions of dollars per year continues.
- The developing nations in **Asia, Latin America and Africa are the richest source** of biodiversity and have enormous wealth of wildlife.

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The cost of elephant tusks can go up to \$ 100 per kg; the leopard fur coat is sold at \$ 100,000 in Japan.

While bird catchers can fetch upto \$ 10,000 for a rare hyacinth macaw, a beautiful coloured bird, from Brazil.

MAN-WILDLIFE CONFLICTS

- Sometimes we come across conflicting situations when wildlife starts causing immense damage and danger to man and under such conditions it becomes very difficult for the forest department to pacify the affected villagers and gain local support for wild-life. Human killing by animals/ leopards/tigers killing human in Sitapur, UP recently (2025).
- Menace of street dogs, pit bull dogs etc. ?

ENDANGERED SPECIES OF INDIA

- **The International Union for Conservation of Nature (IUCN) publishes the Red Data Book which includes the list of endangered species of plants and animals.**

The Wild Life (Protection) Act, of 1972 provides a legal framework for the protection of various species of wild animals and plants, management of their habitats, regulation, and control of trade in wild animals, plants, and products made from them

- **The act also lists schedules of plants and animals that are afforded varying degrees of protection and monitoring by the government.**
- **Schedule 1(Top protection) , Schedule 2 , Schedule 3 species in India.**

In India, nearly 450 plant species have been identified in the categories of endangered, threatened or rare. Existence of about 150 mammals and 150 species of birds is estimated to be threatened while an unknown number of species of insects are endangered.

ENDEMIC SPECIES OF INDIA

Out of about 47,000 species of plants in our country 7000 are endemic. Thus, Indian subcontinent has about 62% endemic flora, restricted mainly to Himalayas, Khasi Hills and Western Ghats.

- Some of the important endemic flora include orchids and species like *Sapria himalayana*, *Uvaria lurida*, *Nepenthes khasiana*, *Pedicularis perroter* etc.

- A large number out of a total of 81,000 species of animals in our country is endemic. The western ghats are particularly rich in amphibians (frogs, toads etc.) and reptiles (lizards, crocodiles etc.). About 62 amphibians and 50% lizards are endemic to Western Ghats.

Different species of monitor lizards (Varanus), reticulated python and Indian Salamander and Viviparous toad Nectophryne are some important endemic species of our country

CONSERVATION OF BIODIVERSITY

- The enormous value of biodiversity due to their genetic, commercial, medical, aesthetic, ecological and optional importance emphasizes the need to conserve biodiversity.
- There are two approaches of biodiversity conservation:
- **(a) In situ conservation (within habitat):** This is achieved by protection of wild flora and fauna in nature itself. e.g. Biosphere Reserves,

Biodiversity Conservation.....

- **National Parks, Sanctuaries, Reserve Forests etc.**
- (b) Ex situ conservation (outside habitats) :** This is done by establishment of gene banks, seed banks, zoos, botanical gardens, culture collections etc.

- **At present we have 7 major Biosphere Reserves, 80 National Parks, 420 wild-life sanctuaries and 120 Botanical gardens in our country covering 4% of the geographic area.**
- **The Biosphere Reserves conserve some representative ecosystems as a whole for long-term in in - situ conservation.**

Biosphere Reserves

- **In India we have Nanda Devi (U.P.), Nokrek (Meghalaya), Manas (Assam), Sunderbans (West Bengal), Gulf of Mannar (Tamil Nadu), Nilgiri (Karnataka, Kerala, Tamil Nadu), Great Nicobars and Similipal (Orisa/Odisha) Biosphere Reserves.**
- **Different Zones : Core - Buffer –Transition Area- Human Settlement-Research Station-Monitoring- Education Training-Tourism**
- **Image in the Next Slide**

The three zones that characterise a Biosphere Reserve are



- Core area
- Buffer zone
- Transition area
- 🏠 Human settlements
- 🎯 Research station
- 📹 Monitoring
- 📖 Education / training
- 📷 Tourism / recreation

- **Within the Biosphere reserves we may have one or more National Parks.**

For example, Nilgiri Biosphere Reserve has two National Parks viz. Bandipur and Nagarhole National Park

- **A National Park is an area dedicated for the conservation of wildlife along with its environment.**

- **It is also meant for enjoyment through tourism but without impairing the environment.**
- **Grazing of domestic animals, all private rights and forestry activities are prohibited within a National Park.**
- **There are Monitoring stations, Research , Training , Human settlement etc. as well.**

Tree transplantation – Curbing deforestation

- For plants, there is one gene sanctuary for Citrus (Lemon family) and one for pitcher plant (an insect eating plant) in Northeast India.
- For the protection and conservation of certain animals, there have been specific projects in our country e.g. Project Tiger, Gir Lion Project, Crocodile Breeding Project, Project Elephant, Snow Leopard Project.

Ex - situ Conservation

- This type of conservation is mainly done for conservation of crop varieties, the wild relatives of crops and all the local varieties with the main objective of conserving the total genetic variability of the crop species for future crop Improvement or afforestation programmes, e.g., even Bonsai can conserve genes. Bonsai are not genetically modified. Any tree can be made bonsai.

Gene bank/seed bank facilities:

- **(i) National Bureau of Plant Genetic Resources (NBPGR) is located in New Delhi. Here agricultural and horticultural crops and their wild relatives are preserved by cryo-preservation of seeds, pollen etc. by using liquid nitrogen at a temperature as low as - 196°C.**

- **Varieties of rice, pearl millet, Brassica, turnip, radish, tomato, onion, carrot, chilli, tobacco, poppy etc. have been preserved successfully in liquid nitrogen for several years without losing seed viability**

- **(ii) National Bureau of Animal Genetic Resources (NBAGR) located at Karnal, Haryana.**
- **It preserves the semen of domesticated bovine animals.**

- **(iii) National Facility for Plant Tissue Culture Repository (NFPTCR) for the development of a facility of conservation of varieties of crop plants/trees by tissue culture. This facility has been created within the NBPGR, New Delhi.**

G 15 Countries

- The G-15 countries have also resolved to set up a network of gene banks to facilitate the conservation of various varieties of aromatic and medicinal plants for which India is the networking co-ordinator country.
- The founder-members were Argentina, Chile, Peru, Brazil, Mexico, Jamaica and Venezuela from the Americas; Egypt, Algeria, Senegal, Nigeria and Zimbabwe from Africa; and India, Malaysia and Indonesia from Asia.
- The first G-15 summit was held in Malaysia in 1990.

Related Material

- Life of earth = 4.6 million years.
- Total number of species on the earth = 2.13 million or 8.7 million ?
- Total number of animals = 1.6 million.
- In the world, some 900 thousand different kinds of living insects are known. This representation approximates 80 percent of the world's species.